



STUDENT'S NAME: _____

TEACHER'S NAME: _____

2022

HURLSTONE AGRICULTURAL HIGH SCHOOL

YEAR 11 ASSESSMENT TASK 3

Mathematics Advanced

General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using a black or blue pen
- NESA approved calculators may be used
- A reference sheet is provided in the Section I booklet
- For questions in Section II, show all relevant mathematical reasoning and/or calculations
- This examination paper is not to be removed from the examination centre

Total marks:
70

Section I – 10 marks (pages 2 – 4)

- Attempt Questions 1 – 10. The multiple-choice answer sheet has been provided
- Allow about 15 minutes for this section

Section II – 60 marks (pages 9– 30)

- Attempt Questions 11 – 15, write your solutions in the spaces provided.
- There are 5 separate question/answer booklets. Extra working pages are available if required.
- Allow about 1 hour and 45 minutes for this section.

SECTION I (10 marks)

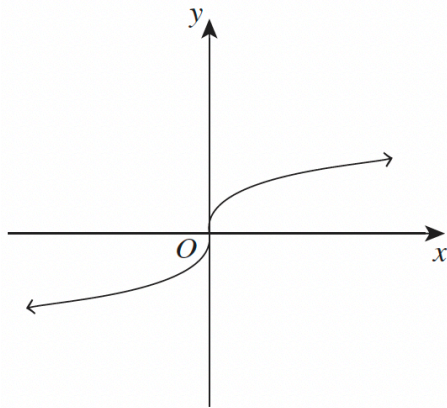
Attempt Questions 1 - 10

Allow about 15 minutes on this section.

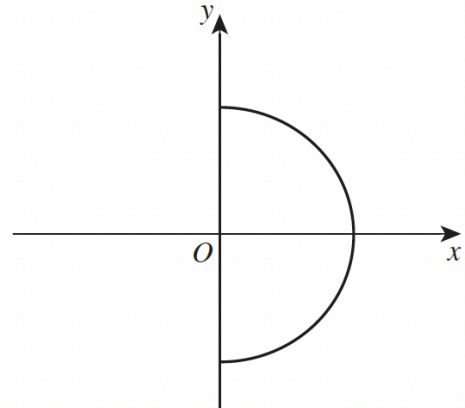
Use the multiple-choice answer sheet provided for Questions 1 – 10.

1. Which one of the following graphs represents a one-to-many relationship?

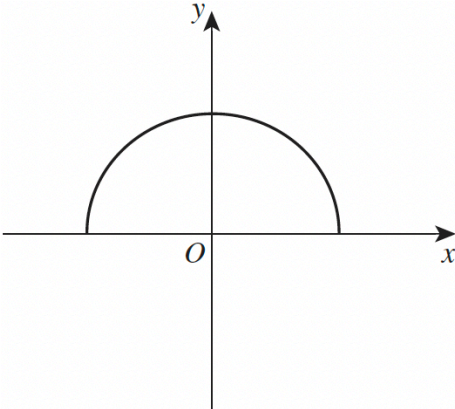
(A)



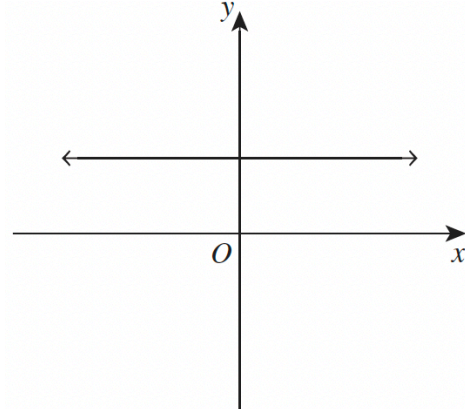
(B)



(C)



(D)



-
2. Which one of the following is **not** a discrete random variable?

- (A) The number of kilometres travelled per litre of petrol.
 - (B) The number of motorists buying petrol at a service station.
 - (C) The number of passengers in a car filling up with petrol at a service station.
 - (D) The number of times per week a motorist needs to refill their car with petrol.
-

3. What is the derivative with respect to x of the expression $\frac{2x^3 - 5x^2 + x}{x}$?

(A) $8x - 30 + \frac{2}{x}$

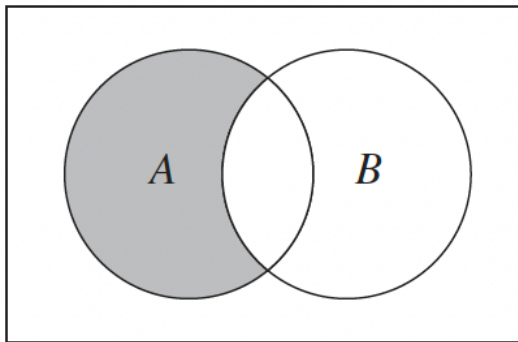
(B) $6x^2 - 10x + 1$

(C) $4x^3 - 5x^2$

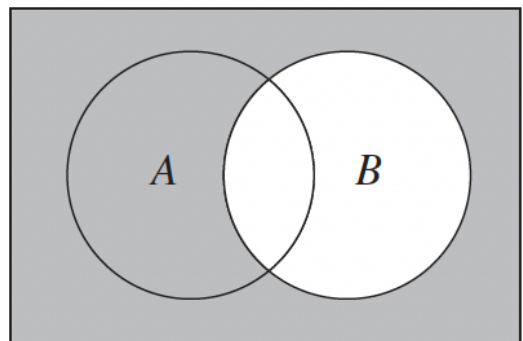
(D) $4x - 5$

4. Which one of the following Venn diagrams represents $A \cup \bar{B}$?

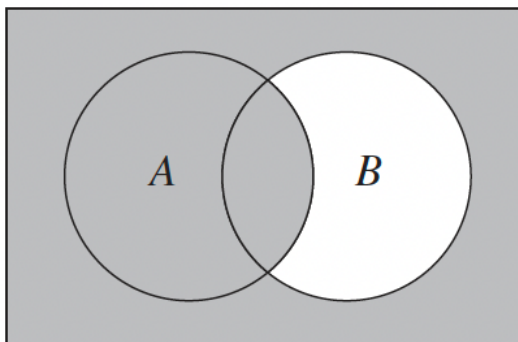
(A)



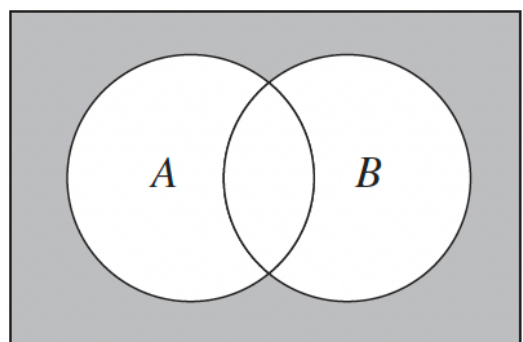
(B)



(C)



(D)



5. Which one of the following sets of ordered pairs represents an odd function?

(A) $\{(2, 3), (3, 2), (5, 7), (7, 5)\}$

(B) $\{(-2, 3), (2, 3), (-5, 7), (7, 5)\}$

(C) $\{(2, 3), (-2, -3), (5, 7), (-5, -7)\}$

(D) $\{(2, 3), (3, 5), (5, 7), (7, 2)\}$

6. Given $\cos \theta = -\frac{8}{17}$ and $\sin \theta$ is also negative, what is the value of $\tan \theta$?

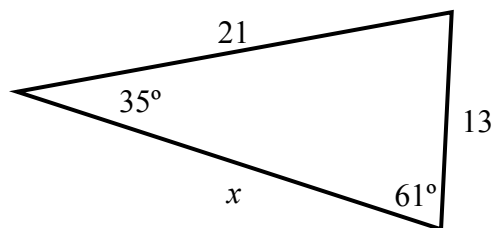
(A) $\frac{8}{15}$

(B) $-\frac{8}{15}$

(C) $-\frac{17}{15}$

(D) $\frac{15}{8}$

7.



NOT TO SCALE
Dimensions are in units.

For the above triangle, which expression will give the correct value for x ?

- (A) $\frac{13\sin 61^\circ}{\sin 35^\circ}$ (B) $13\cos 61^\circ$
(C) $21\sin 84^\circ$ (D) $\sqrt{13^2 + 21^2 - 2 \times 13 \times 21 \times \cos 84^\circ}$

8. What is the angle of inclination to the positive x axis, to the nearest degree, of the tangent to the curve $y = \frac{x^3}{6} + 2x$ at the point where $x = 1$?

- (A) 65° (B) 68°
(C) 79° (D) 112°

9. Over the domain $[0, 2\pi]$, how many solutions are there for the equation $\sin^2 x = 0$?

- (A) 4 (B) 3
(C) 2 (D) 1

10.

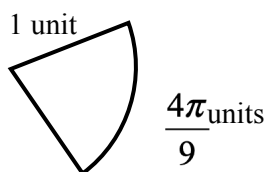


Diagram not to scale.

An arc on the unit circle, of length $\frac{4\pi}{9}$ units, is shown on the diagram above. What is the size of the angle subtended at the centre of the unit circle by this arc in degrees?

- (A) 80° (B) 40°
(C) 20° (D) 10°

End of Section A

**Mathematics Advanced**
Mathematics Extension 1
Mathematics Extension 2**REFERENCE SHEET****Measurement****Length**

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Functions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For $ax^3 + bx^2 + cx + d = 0$:

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\text{and } \alpha\beta\gamma = -\frac{d}{a}$$

Relations

$$(x - h)^2 + (y - k)^2 = r^2$$

Financial Mathematics

$$A = P(1 + r)^n$$

Sequences and series

$$T_n = a + (n - 1)d$$

$$S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(1 - r^n)}{1 - r} = \frac{a(r^n - 1)}{r - 1}, r \neq 1$$

$$S = \frac{a}{1 - r}, |r| < 1$$

Logarithmic and Exponential Functions

$$\log_a a^x = x = a^{\log_a x}$$

$$\log_a x = \frac{\log_b x}{\log_b a}$$

$$a^x = e^{x \ln a}$$

Trigonometric Functions

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab \sin C$$

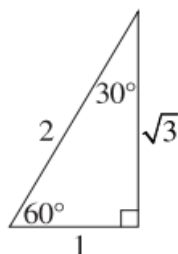
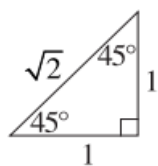
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$l = r\theta$$

$$A = \frac{1}{2}r^2\theta$$



Trigonometric identities

$$\sec A = \frac{1}{\cos A}, \quad \cos A \neq 0$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \quad \sin A \neq 0$$

$$\cot A = \frac{\cos A}{\sin A}, \quad \sin A \neq 0$$

$$\cos^2 x + \sin^2 x = 1$$

Compound angles

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\text{If } t = \tan \frac{A}{2} \text{ then } \sin A = \frac{2t}{1+t^2}$$

$$\cos A = \frac{1-t^2}{1+t^2}$$

$$\tan A = \frac{2t}{1-t^2}$$

$$\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$$

$$\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$$

$$\sin A \cos B = \frac{1}{2}[\sin(A + B) + \sin(A - B)]$$

$$\cos A \sin B = \frac{1}{2}[\sin(A + B) - \sin(A - B)]$$

$$\sin^2 nx = \frac{1}{2}(1 - \cos 2nx)$$

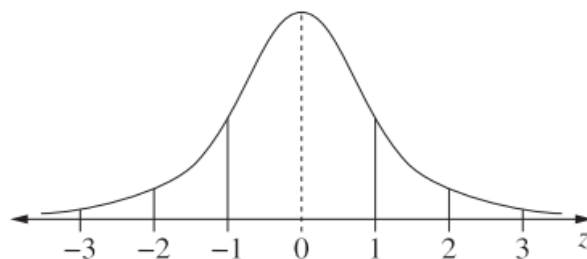
$$\cos^2 nx = \frac{1}{2}(1 + \cos 2nx)$$

Statistical Analysis

$$z = \frac{x - \mu}{\sigma}$$

An outlier is a score
less than $Q_1 - 1.5 \times IQR$
or
more than $Q_3 + 1.5 \times IQR$

Normal distribution



- approximately 68% of scores have z -scores between -1 and 1
- approximately 95% of scores have z -scores between -2 and 2
- approximately 99.7% of scores have z -scores between -3 and 3

$$E(X) = \mu$$

$$\operatorname{Var}(X) = E[(X - \mu)^2] = E(X^2) - \mu^2$$

Probability

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0$$

Continuous random variables

$$P(X \leq r) = \int_a^r f(x) dx$$

$$P(a < X < b) = \int_a^b f(x) dx$$

Binomial distribution

$$P(X = r) = {}^nC_r p^r (1-p)^{n-r}$$

$$X \sim \operatorname{Bin}(n, p)$$

$$\Rightarrow P(X = x)$$

$$= \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \dots, n$$

$$E(X) = np$$

$$\operatorname{Var}(X) = np(1-p)$$

Differential Calculus

Function

Derivative

$$y = f(x)^n$$

$$\frac{dy}{dx} = n f'(x) [f(x)]^{n-1}$$

$$y = uv$$

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$y = g(u) \text{ where } u = f(x)$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$y = \frac{u}{v}$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$y = \sin f(x)$$

$$\frac{dy}{dx} = f'(x) \cos f(x)$$

$$y = \cos f(x)$$

$$\frac{dy}{dx} = -f'(x) \sin f(x)$$

$$y = \tan f(x)$$

$$\frac{dy}{dx} = f'(x) \sec^2 f(x)$$

$$y = e^{f(x)}$$

$$\frac{dy}{dx} = f'(x) e^{f(x)}$$

$$y = \ln f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$$

$$y = a^{f(x)}$$

$$\frac{dy}{dx} = (\ln a) f'(x) a^{f(x)}$$

$$y = \log_a f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{(\ln a) f(x)}$$

$$y = \sin^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \cos^{-1} f(x)$$

$$\frac{dy}{dx} = -\frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \tan^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{1 + [f(x)]^2}$$

Integral Calculus

$$\int f'(x) [f(x)]^n dx = \frac{1}{n+1} [f(x)]^{n+1} + c$$

where $n \neq -1$

$$\int f'(x) \sin f(x) dx = -\cos f(x) + c$$

$$\int f'(x) \cos f(x) dx = \sin f(x) + c$$

$$\int f'(x) \sec^2 f(x) dx = \tan f(x) + c$$

$$\int f'(x) e^{f(x)} dx = e^{f(x)} + c$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + c$$

$$\int f'(x) a^{f(x)} dx = \frac{a^{f(x)}}{\ln a} + c$$

$$\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}} dx = \sin^{-1} \frac{f(x)}{a} + c$$

$$\int \frac{f'(x)}{a^2 + [f(x)]^2} dx = \frac{1}{a} \tan^{-1} \frac{f(x)}{a} + c$$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\int_a^b f(x) dx$$

$$\approx \frac{b-a}{2n} \left\{ f(a) + f(b) + 2[f(x_1) + \dots + f(x_{n-1})] \right\}$$

where $a = x_0$ and $b = x_n$

Combinatorics

$${}^nP_r = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}^nC_r = \frac{n!}{r!(n-r)!}$$

$$(x+a)^n = x^n + \binom{n}{1}x^{n-1}a + \cdots + \binom{n}{r}x^{n-r}a^r + \cdots + a^n$$

Vectors

$$|\underline{u}| = |x_1\underline{i} + y_1\underline{j}| = \sqrt{x^2 + y^2}$$

$$\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos \theta = x_1x_2 + y_1y_2,$$

$$\text{where } \underline{u} = x_1\underline{i} + y_1\underline{j}$$

$$\text{and } \underline{v} = x_2\underline{i} + y_2\underline{j}$$

$$\underline{r} = \underline{a} + \lambda \underline{b}$$

Complex Numbers

$$\begin{aligned} z = a + ib &= r(\cos \theta + i \sin \theta) \\ &= re^{i\theta} \end{aligned}$$

$$\begin{aligned} [r(\cos \theta + i \sin \theta)]^n &= r^n(\cos n\theta + i \sin n\theta) \\ &= r^n e^{in\theta} \end{aligned}$$

Mechanics

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

$$x = a \cos(nt + \alpha) + c$$

$$x = a \sin(nt + \alpha) + c$$

$$\ddot{x} = -n^2(x - c)$$

SECTION II

Name: _____

70 marks**Attempt Questions 11 - 15****Allow about 1 hours and 45 minutes on this section.**

Answer each question in the spaces provided. Extra working space is available after each question. If you need to use this extra space, you must clearly indicate this in the main solution space, and then clearly indicate the question number and part that you are answering in the extra space.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

2022 Year 11 Mathematics Advanced Task 3 Examination Section II**Question 11 (12 marks)****Marks**

- (a) Differentiate the expression $(x^2 - 5)(9x - 2x^3)$ with respect to x .

2

- (b) Differentiate $f(x) = x^2 + 5x$ using $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$.

2

(c) Find $\frac{dy}{dx}$ if $y = (2x^2 - 3)^5$.

2

(d) Find the exact gradient of the tangent to the curve $y = \frac{5x+2}{3x+1}$ when $x = 1$.

3

(e) Find the equation of the normal to the curve $y = x^2 - 4x + 4$ at the point $(3, 1)$.

HAHS 2022 Year 11 Mathematics Advanced Yearly Examination (Assessment Task 3)

Section II Extra writing space

If you use this space, clearly indicate which question you are answering.

2022 Year 11 Mathematics Advanced Task 3 Examination Section II

Name: _____

Question 12 (12 marks)

Marks

- (a) Find the domain of the function $f(x) = \frac{x}{x-3} + \sqrt{x}$.

2

- (b) Determine whether the function $f(x) = \frac{1}{x^3 - 4x}$ is odd, even or neither.
Justify your answer.

2

- (c) (i) Show that $\frac{2x^2 + x - 1}{x^2 - 1} = 2 + \frac{1}{x-1}$.

2

2

1

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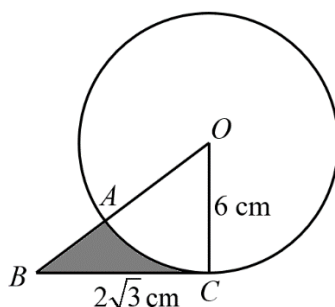
If you use this space, clearly indicate which question you are answering.

Name: _____

Question 13 (12 marks)

Marks

- (a) The diagram shows the circle with centre O . OAB is a straight line, BC is the tangent at C with length $2\sqrt{3}$ cm and radius 6 cm.

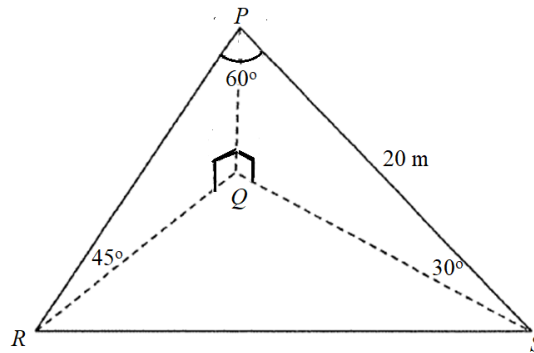


- (i) Show $\angle BOC$ is $\frac{\pi}{6}$. 1

- (ii) Find the area of sector OAC in terms of π . 1

- (iii) Hence, find the area of the shaded region.
(Give your answer correct to 2 decimal places.) 2

- (b) The diagram shows R, S, Q are three points on the same horizontal ground. PQ is a vertical pole on the plane RSQ . PR and PS are ropes with $PS = 20$ m and $\angle RPS = 60^\circ$.



- (i) Show that PQ is 10 m.

1

- (ii) Find PR , giving your answer as an exact value.

1

- (iii) Find the distance between the points R and S .
(Give your answer correct to 1 decimal place.)

2

(iv) Find the size of $\angle RQS$ correct to the nearest degree.

3

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(v) Given that R is due south of Q, find the bearing of Q from S to the nearest degree.

1

[illegible]

End of Question 13

Section II Extra writing space

If you use this space, clearly indicate which question you are answering.

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2022 Year 11 Mathematics Advanced Task 3 Examination Section II

Name: _____

Question 14 (12 marks)

Marks

(a) Solve:

2

$$2\cos\theta = -\sqrt{3} \text{ for } -\pi \leq \theta \leq \pi.$$

(b) For what values of x is $y = \tan x$ undefined for $0^\circ \leq x \leq 720^\circ$?

2

(c) Prove the identity:

3

$$\frac{\sin\theta}{1+\cos\theta} + \frac{1+\cos\theta}{\sin\theta} = 2\operatorname{cosec}\theta$$

(d) Solve: $2\cos^2 2x = 6\sin^2 2x, 0^\circ \leq x \leq 90^\circ$

3

(e) Draw a neat sketch of the graph of $y = -4\cos(2x), 0 \leq x \leq 2\pi$.

2

End of Question 14

Section II Extra writing space

If you use this space, clearly indicate which question you are answering.

Section II Extra writing space

If you use this space, clearly indicate which question you are answering.

Name: _____

Question 15 (12 marks)

Marks

(a) A game is played where a person rolls a fair, six-sided die. If the die lands on the number 2, the person playing wins and the game ends.

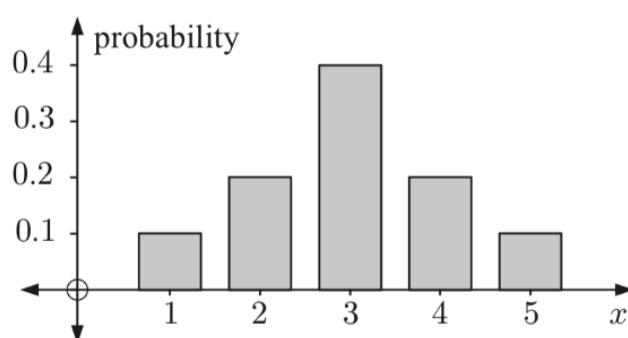
(i) By using a probability tree diagram or otherwise, calculate the probability that the game ends after the third roll.

2

(ii) Calculate the probability that the person rolls the die no more than three times in a single game.

1

- (b) A discrete random variable X has the probability distribution shown below.



- (i) Find $E(X)$.

1

- (ii) Find the standard deviation.

2

- (iii) Find $P(x > \mu + \sigma)$.

1

- (c) The probability that Michael will score *more than* 100 points in a game of bowling is $\frac{31}{40}$. 1

A commentator states that the probability that Michael will score *less than* 100 points in a game of bowling is $\frac{9}{40}$.

Is the commentator correct? Give a reason for your answer.

- (d) The events X and Y are independent. Given that $P(X) = 0.3$ and $P(Y) = 0.42$, find $P(X \cup Y)$. 2

- (e) A discrete random variable X has the probability distribution shown below.

2

Y	-1	0	1	2
$P(Y = y)$	0.1	a	0.3	b

Find a and b if $E(X) = 0.9$.

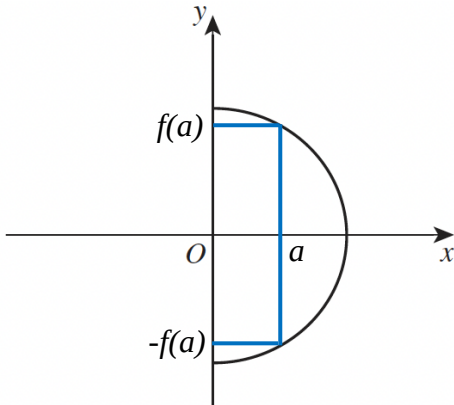
End of Question 15

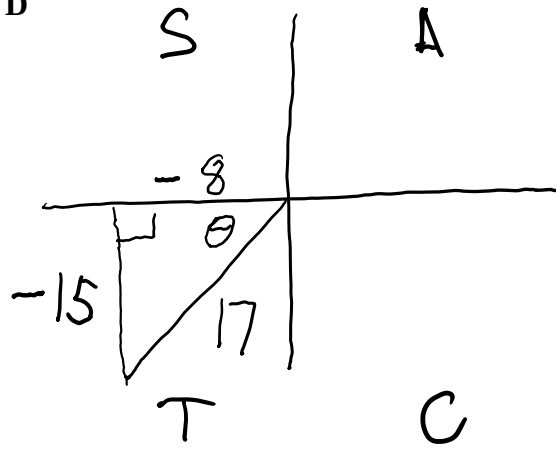
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Section II Extra writing space

If you use this space, clearly indicate which question you are answering.

Year 11 Advanced Mathematics Yearly Examination (Task 3) 2022		
Question No. 1 – 10 (Multiple Choice) Solutions and Marking Guidelines		
Outcomes Addressed in this Question		
MA11-1	uses algebraic and graphical techniques to solve, and where appropriate, compare alternative solutions to problems	
MA11-2	uses the concepts of functions and relations to model, analyse and solve practical problems	
MA11-3	uses the concepts and techniques of trigonometry in the solution of equations and problems involving geometric shapes	
MA11-4	uses the concepts and techniques of periodic functions in the solutions of trigonometric equations or proof of trigonometric identities	
MA11-5	interprets the meaning of the derivative, determines the derivative of functions and applies these to solve simple practical problems	
MA11-7	uses concepts and techniques from probability to present and interpret data and solve problems in a variety of contexts, including the use of probability distributions	
Outcome	Solutions	Marking Guidelines
MA11-2	1. B  From the diagram for B it can be seen that one value of x, in this case a, will map to many function values, in this case $f(a)$ and $-f(a)$.	1 mark Correct Answer
MA11-7	2. A The number of kilometres travelled is not measured in discrete units. The number can be measured to the desired level of accuracy, hence it is a continuous variable. The number of motorists (B), the number of passengers (C) and number of times per week (D) can only be measured as discrete variables as in each case the variables can only be measured as whole numbers.	1 mark Correct Answer
MA11-5	3. D As the expression is a fraction with a monomial denominator, simplify the fraction before differentiation. $\frac{2x^3 - 5x^2 + x}{x} = 2x^2 - 5x + 1$ $\frac{d}{dx}(2x^2 - 5x + 1) = 4x - 5$	1 mark Correct Answer

MA11-7	4. A $A \cap \bar{B}$ means we need to shade everything in set A that overlaps with everything that is <i>not</i> in B, hence the answer A. Answer B represents $\bar{B}$. Answer C represents $A \cup \bar{B}$. Answer D represents $\overline{(A \cup B)}$	1 mark Correct Answer
MA11-2	5. C For an odd function $f(x) = -f(-x)$ Since the sets consist of individual points, if we have the point (2, 3) the set must also contain (-2, -3). If the set contains (5, 7), it must also contain (-5, -7). Answer C shows the only set with this property.	1 mark Correct Answer
MA11-3	6. D  From the given information the angle θ must be in the third quadrant as both sine and cosine are negative. Constructing the diagram above using the given cosine ratio and Pythagoras' Theorem we can then see that $\tan \theta = \frac{15}{8}$.	1 mark Correct Answer
MA11-3	7. D Deducing that the missing angle in the triangle is 84° (using angle sum of a triangle), we can then apply the cosine rule to find an unknown side: $x = \sqrt{13^2 + 21^2 - 2 \times 13 \times 21 \times \cos 84^\circ}$	1 mark Correct Answer

MA11-5	8. B $y = \frac{x^3}{6} + 2x$ $\frac{dy}{dx} = \frac{3x^2}{6} + 2$ $= \frac{x^2}{2} + 2$ The tangent ratio of the angle of inclination is the same as the gradient of the curve. At $x = 1$, $\frac{dy}{dx} = \frac{1}{2} + 2 = 2.5$ ie. $\tan \theta = 2.5$ $\therefore \theta \approx 68^\circ$	1 mark Correct Answer
MA11-4	9. B $\sin^2 x = 0$ $\therefore \sin x = 0$ $x = 0, \pi, 2\pi \quad [0, 2\pi]$ ie. 3 solutions	1 mark Correct Answer
MA11-4	10. A On the unit circle, an arc of length 1 unit subtends an arc of 1 radian. It then follows that an arc of length $\frac{4\pi}{9}$ units would subtend an angle of $\frac{4\pi}{9}$ radians. Converting this angle to degrees: $\frac{4\pi}{9} \times \frac{180}{\pi} = 80^\circ$	1 mark Correct Answer
MA11-1	This outcome is generally assessed in all questions in this section.	

Year 11 Advanced Mathematics Yearly Examination (Task 3) 2022		
Question No. Solutions and Marking Guidelines		
Outcomes Addressed in this Question		
For example: MA11-5 interprets the meaning of the derivative, determines the derivative of functions and applies these to solve simple practical problems		
Outcome	Solutions	Marking Guidelines
MA11-5	Question 11 a) $\frac{dy}{dx} = (x^2 - 5)(9 - 6x^2) + (9x - 2x^3)(2x)$ Or $\frac{dy}{dx} = -10x^4 + 57x^2 - 45$	2 marks Correct solution, giving correct derivative of function. 1 mark Some progress towards a correct solution.
	b) $f'(x) = \lim_{h \rightarrow 0} \frac{x^2 + 2hx + h^2 + 5x + 5h - x^2 - 5x}{h}$ $= \lim_{h \rightarrow 0} \frac{2hx + h^2 + 5h}{h}$ $= \lim_{h \rightarrow 0} \frac{h(2x + h + 5)}{h}$ $= \lim_{h \rightarrow 0} 2x + h + 5$ $= 2x + 5$	2 marks Correct solution, giving correct substitution into $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \text{ and}$ correct solution $f'(x) = 2x + 5.$ 1 mark Some progress towards a correct solution, giving correct substitution into $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$
MA11-5	c) $\frac{dy}{dx} = 5(2x^2 - 3)(4x)$ Or $\frac{dy}{dx} = (20x)(2x^2 - 3)$	2 marks Correct solution, giving correct derivative of function. 1 mark Some progress towards a correct solution.
MA11-5	d) $\frac{dy}{dx} = \frac{(3x+1)(5) + (5x+2)(3)}{(3x+1)^2}$ Or $\frac{dy}{dx} = \frac{-1}{(3x+1)^2}$ At $x = 1$ $\frac{dy}{dx} = \frac{-1}{(3(1)+1)^2}$ $= -\frac{1}{16}$ $\therefore$ gradient of tangent $= -\frac{1}{16}$ when $x = 1$	3 marks Correct solution, giving correct derivative of function and finding $m = -\frac{1}{16}.$ 2 marks Some progress towards correct solution, giving correct derivative of function. 1 mark Some progress towards a correct solution.

MA11-5

e)

$$\frac{dy}{dx} = 2x - 4$$

$$\text{at } x = 3, \frac{dy}{dx} = 2(3) - 4 = 2$$

$\therefore$ gradient of tangent to curve at $x = 3$ is $m_T = 2$

$\therefore$ gradient of normal to curve at $x = 3$ is $m_N = -\frac{1}{2}$

$\therefore$ Equation of normal is:

$$y - 1 = -\frac{1}{2}(x - 3)$$

$$x + 2y - 5 = 0$$

3 marks

Correct solution, giving correct derivative of function, finding gradient of normal at (3,1) and equation of normal at (3,1).

2 marks

Some progress towards correct solution, giving correct derivative of function and finding gradient of normal at (3,1).

1 mark

Some progress towards a correct solution, giving correct derivative of function and finding gradient of tangent.

Year 11 Advanced Mathematics Yearly Examination (Task 3) 2022		
Question No. 12		Solutions and Marking Guidelines
Outcomes Addressed in this Question		
MA11-1	uses algebraic and graphical techniques to solve, and where appropriate, compare alternative solutions to problems	
MA11-2	uses the concepts of functions and relations to model, analyse and solve practical problems	
Outcome	Solutions	Marking Guidelines
	For example: (a) D: $x \geq 0, x \neq 3$ <i>alternative notation:</i> $[0, 3) \cup (3, \infty)$ (b) $f(x) = \frac{1}{x^3 - 4x}$ $f(-x) = \frac{1}{(-x)^3 - 4(-x)}$ $= \frac{1}{-x^3 + 4x}$ $= -\frac{1}{x^3 - 4x}$ $f(-x) = -f(x)$ $\therefore$ odd function (c)(i) $\text{LHS} = \frac{2x^2 + x - 1}{x^2 - 1}$ $= \frac{(2x-1)(x+1)}{(x-1)(x+1)}$ $= \frac{2x-1}{x-1}$ $= \frac{2x-2+1}{x-1} \quad \left(\begin{array}{l} \text{this is a key step} \\ \text{to be shown} \end{array} \right)$ $= \frac{2(x-1)+1}{x-1}$ $= 2 + \frac{1}{x-1}$ $= \text{RHS}$	2 marks Correct solution 1 mark Substantial progress towards correct solution. 2 marks Correct solution 1 mark Substantial progress towards correct solution. 2 marks Correct solution 1 mark Substantial progress towards correct solution.

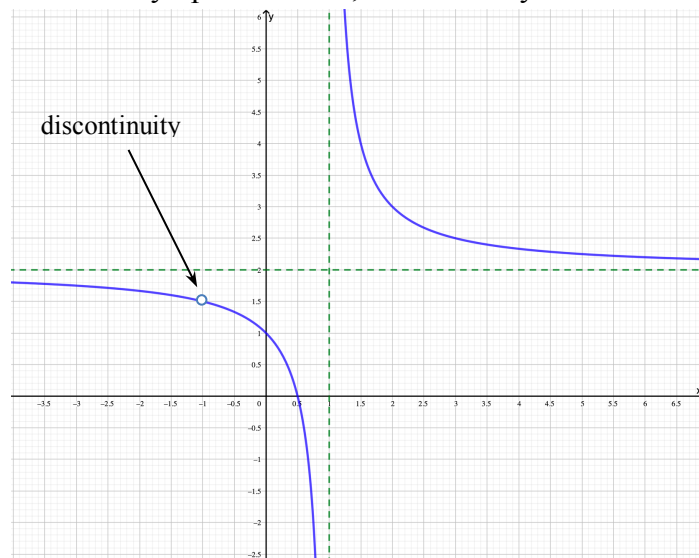
Question 12 continued...

(c)(ii) Note:

$$D: x^2 - 1 \neq 0$$

$$\therefore x \neq \pm 1$$

$\Rightarrow$ asymptote at $x = 1$, discontinuity at $x = -1$



3 marks

Correct solution, giving correct asymptotes and intercepts.

Note: open circle at $(-1, 1.5)$ was not required for full marks in this question... this will not always be the case

2 marks

Substantially correct solution.

1 mark

Some progress towards a correct solution.

(d)(i) $f(x) = x^2 - 9$

$$f(g(x)) = [g(x)]^2 - 9$$

$$= [3x + 1]^2 - 9$$

$$= 9x^2 + 6x + 1 - 9$$

$$= 9x^2 + 6x - 8$$

2 marks

Correct solution

1 mark

Substantial progress towards correct solution.

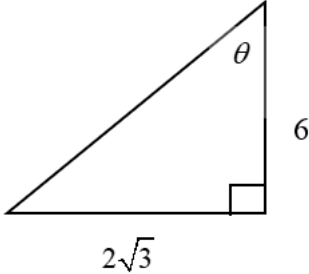
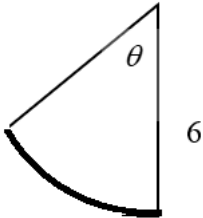
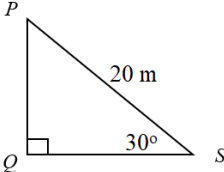
(d)(ii) $h(2) = 9(2)^2 + 6(2) - 8$

$$= 36 + 12 - 8$$

$$= 40$$

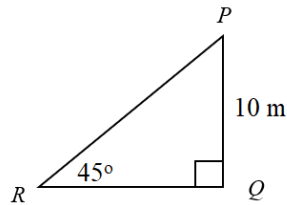
1 mark

Correct solution

Year 11 Advanced Mathematics Yearly Examination (Task 3) 2022		
Question No. 13 Solutions and Marking Guidelines		
Outcomes Addressed in this Question		
MA11-3 uses the concepts and techniques of trigonometry in the solution of equations and problems involving geometric shapes		
Outcome	Solutions	Marking Guidelines
MA11-3	(a)(i) $\tan \angle BOC = \frac{2\sqrt{3}}{6}$ $= \frac{\sqrt{3}}{3}$ $\angle BOC = 30^\circ \times \frac{\pi}{180^\circ}$ $= \frac{\pi}{6}$ 	1 mark Correct solution
	(ii) Area of sector $= \frac{1}{2} r^2 \theta$ $= \frac{1}{2} \times 6^2 \times \frac{\pi}{6}$ $= 3\pi \text{ cm}^2$ 	1 mark Correct solution
	(iii) Area of shaded region = $\frac{1}{2} \times \text{base} \times \text{height} - \text{area of sector}$ $= \frac{1}{2} \times 2\sqrt{3} \times 6 - 3\pi$ $= 0.97 \text{ cm}^2$	2 marks Correct solution 1 mark Substantial progress towards correct solution. or Some progress towards a correct solution.
	(b)(i) In $\triangle PQS$ $\sin 30^\circ = \frac{PQ}{20}$ $PQ = 20 \sin 30^\circ$ $= 10 \text{ m}$ 	1 mark Correct solution

(ii) In $\triangle PQR$

$$\begin{aligned}\sin 45^\circ &= \frac{PQ}{PR} \\ PR &= \frac{10}{\sin 45^\circ} \\ &= 10\sqrt{2} \text{ m}\end{aligned}$$



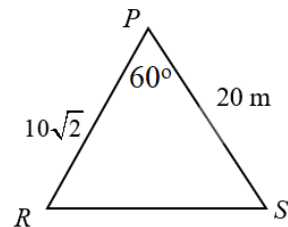
(iii) In $\triangle PRS$

$$RS^2 = (10\sqrt{2})^2 + 20^2 - 2(10\sqrt{2})(20)\cos 60^\circ \quad (\text{Not a right angled triangle})$$

$$\begin{aligned}&= 200 + 400 - 200\sqrt{2} \\ &= 317.1573\end{aligned}$$

$$\begin{aligned}RS &= \sqrt{317.1573} \\ &= 17.8 \text{ m}\end{aligned}$$

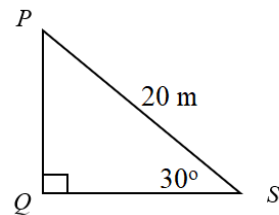
$\therefore$ The distance RS is 17.8 m.



(iv) In $\triangle PQS$

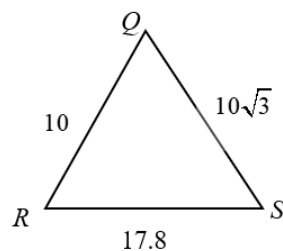
$$\begin{aligned}\cos 30^\circ &= \frac{QS}{PS} \\ QS &= 20 \cos 30^\circ \\ &= 10\sqrt{3} \text{ m}\end{aligned}$$

$$PQ = QR = 10 \text{ m}$$



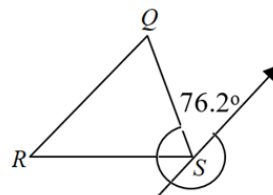
In $\triangle QRS$

$$\begin{aligned}\cos \angle RQS &= \frac{10^2 + (10\sqrt{3})^2 - 17.8^2}{2(10)(10\sqrt{3})} \\ &= \frac{400 - 17.8^2}{200\sqrt{3}} \\ \angle RQS &= 76.2^\circ\end{aligned}$$



(v) Bearing

$$\begin{aligned}&= 360^\circ - 76.2^\circ \\ &= 284^\circ\end{aligned}$$



1 mark

Correct solution

2 marks

Correct solution

1 mark

Substantial progress towards correct solution.

or

Some progress towards a correct solution.

3 marks

Correct solution

2 marks

Substantial progress towards correct solution.

1 mark

Some progress towards a correct solution.

1 mark

Correct solution

Year 11 Advanced Mathematics Yearly Examination (Task 3) 2022		
Question No. 14 Solutions and Marking Guidelines		
Outcomes Addressed in this Question		
MA11-4 uses the concepts and techniques of periodic functions in the solutions of trigonometric equations or proof of trigonometric identities		
Question	Solutions	Marking Guidelines
(a)	$\cos\theta = -\frac{\sqrt{3}}{2}$ $\text{Related angle} = \frac{\pi}{6}$ $\theta = \pi - \frac{\pi}{6}, \pi + \frac{\pi}{6}$ $\theta = \frac{5\pi}{6}, -\frac{5\pi}{6}$	2 marks Correct solution (in radians) within the given range. 1 mark Solution with incorrect range used OR in degrees OR substantial progress towards the correct solution.
(b)	$y = \tan x$ is undefined for: $x = 90^\circ, 270^\circ, 360^\circ + 90^\circ, 720^\circ - 90^\circ$ $x = 90^\circ, 270^\circ, 450^\circ, 630^\circ$	2 marks Provides the correct solution in degrees. 1 mark Provides any TWO values of x in degrees
(c)	$\begin{aligned} \text{LHS} &= \frac{\sin\theta}{1 + \cos\theta} + \frac{1 + \cos\theta}{\sin\theta} \\ &= \frac{\sin^2\theta + (1 + \cos\theta)^2}{\sin\theta(1 + \cos\theta)} \\ &= \frac{\sin^2\theta + 1 + 2\cos\theta + \cos^2\theta}{\sin\theta(1 + \cos\theta)} \\ &= \frac{2 + 2\cos\theta}{\sin\theta(1 + \cos\theta)} \\ &= \frac{2(1 + \cos\theta)}{\sin\theta(1 + \cos\theta)} \\ &= \frac{2}{\sin\theta} \\ &= 2\operatorname{cosec}\theta \\ &= \text{RHS} \end{aligned}$	3 Marks Provides the correct solution. 2 Marks Shows substantial progress by simplifying the numerator. 1 Mark Writes the expression under a common denominator

(d)

$$2\cos^2(2x) - 6\sin^2(2x) = 0$$

$$2(\cos^2(2x) - 3\sin^2(2x)) = 0$$

$$2(\cos(2x) + \sqrt{3}\sin(2x))(\cos(2x) - \sqrt{3}\sin(2x)) = 0$$

$$\cos(2x) + \sqrt{3}\sin(2x) = 0$$

or

$$\cos(2x) - \sqrt{3}\sin(2x) = 0$$

$$\tan(2x) = \frac{1}{\sqrt{3}} \text{ or } \tan(2x) = -\frac{1}{\sqrt{3}}$$

$$x = 75^\circ \text{ or } x = 15^\circ$$

3 Marks

Provides both solutions AND correct factorisation of trigonometric function.

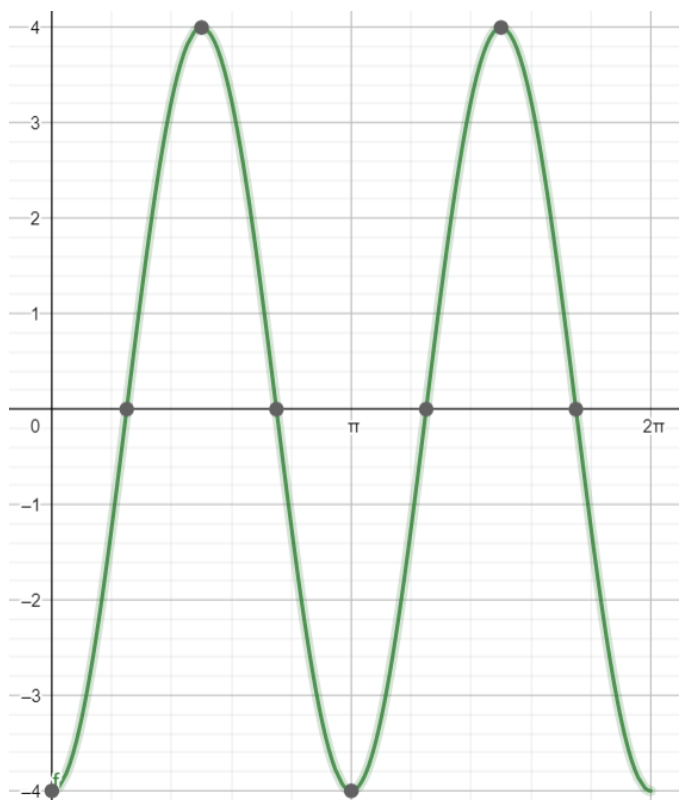
2 Marks

Provides correct factorisation AND one other correct solution.

1 Mark

Provides one correct solution OR correct factorisation.

(e)

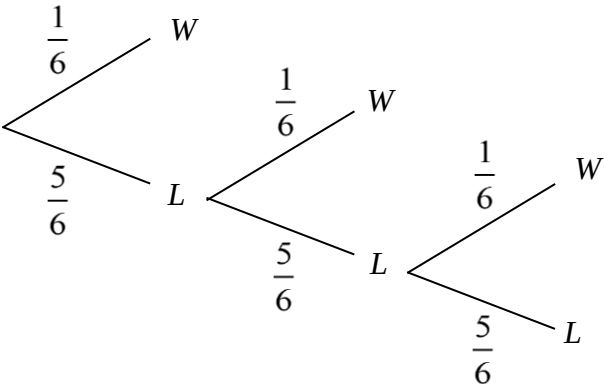


2 Marks

Sketch is correct and neat with key intercept points clearly labelled.

1 Mark

Shape of the sketch is correct, however, messy or missing key intercept points.

Year 11 Question No.15	Mathematics Advanced Solutions and Marking Guidelines	Task 3	Examination 2022
MA11- 7	uses concepts and techniques from probability to present and interpret data and solve problems in a variety of contexts, including the use of probability distributions		
Part	Solutions	Marking Guidelines	
(a)	i)  $P(LLW) = \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} = \frac{25}{216}$	2 marks Correct solution 1 mark Single error	
	ii) $P(W) + P(LW) + P(LLW) = \frac{1}{6} + \frac{5}{6} \times \frac{1}{6} + \frac{25}{216}$ $= \frac{91}{216}$	1 mark Correct solution	
	i) $E(X) = \sum f(x)x$ $= 0.1 \times 1 + 0.2 \times 2 + 0.4 \times 3 + 0.2 \times 4 + 0.1 \times 5$ $= 3$ ii) $V(X) = E(X^2) - E(X)^2$ $= \sum f(x)^2 x^2 - 3^2$ $= 0.1^2 \times 1 + 0.2^2 \times 2 + 0.4^2 \times 3 + 0.2^2 \times 4 + 0.1^2 \times 5 - 3^2$ $= 1.2$ $s = \sqrt{V(X)}$ $= 1.095$	1 mark Correct Solution 2 marks Correct solution 1 mark Single error	
	iii) $P(x > m + s) = P(x > 3 + 1.095)$ $= P(x > 4.095)$ $= P(x = 5)$ $= 0.1$	1 mark Correct solution	

(c)	No, since the commentator did not consider the probability of a score of exactly 100; it is not reasonable to assume it has a chance of 0%.	1 mark Correct conclusion with correct justification
(d)	If independent $P(X \cap Y) = P(X)P(Y)$ $= 0.3 \times 0.42$ $= 0.126$ $P(X \cup Y) = P(X) + P(Y) - P(X \cap Y)$ $= 0.3 + 0.42 - 0.126$ $= 0.594$ NOTE: students should revise the difference between independent events and mutually exclusive events.	2 marks Correct solution 1 mark Significant progress that would lead to a correct answer
(e)	$E(X) = \sum f(x)x$ $0.9 = 0.1 \times (-1) + a \times 0 + 0.3 \times 1 + b \times 2$ $b = 0.35$ $\sum f(x) = 1$ $0.1 + a + 0.3 + b = 1$ $0.1 + a + 0.3 + 0.35 = 1$ $a = 0.25$	2 marks Correct solution 1 mark Significant progress that would lead to a correct answer